

Customer:	Sales Order:	Invoice No.:
Industrial Piping Specialists	0630005397	1600051929
PO Box 681270	PO:	Date of Issue:
Tulsa,OK,74158	WP012766	02/14/2025
918-437-9100		
Certificate No.:	1600051929-2	
Product Description:	5L 8.625 0.322 28.58# X52BDRH5BW-20P2	
Specification:	5L 46th Edition May 2019	

Analysis (H/P)	Chemical Composition (Max %)																	
	C	Si	Mn	P	S	Ni	Cr	Mo	Cu	V	Al	Nb	Ti	B	Ca	N	CEp _{cm} (API Max. A.C.)	CE Results
H	0.06	0.210	0.80	0.013	0.002	0.05	0.06	0.020	0.13	0.002	0.024	0.028	0.015	0.0003	0.0020	0.0062	0.25	0.12
P	0.06	0.222	0.80	0.013	0.001	0.05	0.06	0.010	0.13	0.004	0.020	0.025	0.015	0.0003	0.0022	0.0075	0.25	0.12
P	0.06	0.220	0.80	0.013	0.001	0.05	0.06	0.010	0.13	0.004	0.020	0.025	0.015	0.0004	0.0020	0.0076	0.25	0.12

Remarks:
 1: This material meets: Grades ASTM A53/A53M-22 Grade B
 API 5L HFW PSL2 X52M,
 2: This product meets the requirements of ASME BPVC.II.A Specification SA-53B-2023
 3: Made and Melted in USA
 4: Date of Manufacture 02/2025
 5: This product is silicon and/or aluminum fully killed,
 6: Certificate meets the requirements of ISO 10474:2013 (Inspection Certificate 3.1.B)
 7: This material meets the requirements of NACE MR0175-2021 / ISO 15156:2020

We hereby certify that the material herein has been made and tested in accordance with the above specification and also with the requirement called for by the above order.

Quality Manager

Globe X-Ray Services, Inc.
8441 S. Union Ave.
Tulsa, OK 74132
918-446-1696

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Certification # 194846

Magnetic Particle Examination Record

Customer: Tulsa Tube Bending Date: 6/12/2026

Job/P.O. No.: 109149 Reference Code/Specification: ASME SEC V

MT Procedure No. GXS IV - C Rev. No. 1/26 Acceptances: ASME SEC VIII Div. I

Magnetizing Equipment: Parker Contour Probe B300 Serial No. 1861 Technique: X Yoke Prods

Current Type: X A/C D/C Magnetic Particle Type: Wet X Dry Fluorescent X Visible

Amperage (Prod Technique Only) Amps/inch of Prod Spacing.

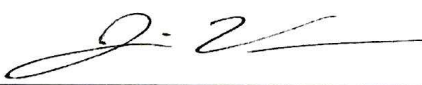
Part/Item Material Type: Carbon Material Thickness: Varies

Lighting Equipment: Fluorescent Light Model # Serial #
X White Light using Parker Research Y300 or Y400 Probe light
 Natural White Light
 Other:

Light Intensity on surface to be examined: 100 Min foot candles N/A $\mu\text{W}/\text{cm}^2$

Comments: Test The Entire Extrados (Back Half) of the Bend. From N.A to N.A, Including Weld Seam
Verify Residual Magnetism at Each End is <25 Gauss

Examination Results				
Qty:	Part Identification	Accept	Reject	Indication Description
4	8.625" O.D. X 0.322" Wall, 40" CLR, API5L X52 ERW Pipe HT# B416745 Bend ID # 109149-1-1, 109149-1-2, 109149-1-3, 109149-1-4,	X		None
2	8.625" O.D. X 0.322" Wall, 40" CLR, API5L X52 ERW Pipe HT# B416740 Bend ID # 109149-2-1, 109149-2-2	X		None

Examine:  Jaime Vera Qualification Level: II

GLOBE X-RAY ASSUMES NO RESPONSIBILITY FOR THE INTEGRITY OF THE STRUCTURE OR FOR ANY LOSSES OF
ANY KIND DUE TO ANY KIND OF INTERPRETATION



Tulsa Tube Bending
4192 S. Galveston Ave.
Tulsa, OK 74107
Tel: 918-446-4461
Fax: 918-446-1595

Serving others. Building people. Pursuing excellence.

Certificate of Conformance

Customer: DNOW Certificate No.: 20260615

Customer PO No.: 662974020 Issue Date: 6/15/2026

TTB Job No.: 26H109149

Material Size, Grade: 8.63 inch OD x 0.322 inch w.t., API-5L-X52 PSL 2 ERW Quantity of Bends: 6

Material Heat No(s): B416745

Bend ID Nos.: 109149-1-1 109149-1-2 109149-1-3 109149-1-4

Material Heat No(s): B416740

Bend ID Nos.: 109149-2-2 109149-2-1

We hereby certify that the aforementioned induction bends have been inspected and conform to all of the requirements of the Purchase Order. The induction bends have been manufactured in conformance with:

ASME B16.49-2023

*With the following notes and exceptions:

* B16.49 SR15.1 - Bends will be in the as bent condition in accordance with ASME B16.49 SR15.1.

* SR15.8.2 Multiple heats of material used for production bends based on similar chemistries.

* Segmentable per SR15.3, 1% max ovality in the bend area.

Approved By: Nate Wallace Date: 6/15/2026



20701 FM 521 ROSSARON, TX 77563
+1-713-991-5670

COATING INSPECTION RECORD

KF-010 FBE 5/27/22 REV2

CUSTOMER: TTB		DATE: 7/7/2026		ITEMS: 6 PCS	
CUSTOMER PO: 26H 109149		PAINT SPEC: TPE 26907SP-CONSUMER ENERGY			
SURFACE PREPARATION					
ITEM SALT TEST	N/A ☐	RESULT: <1	SSPC SP-1 IF NEEDED	DATE	7/7/2026
ABRASIVE SALT TEST	N/A ☐	RESULT: <1	CLEANLINESS REQUIRED	TIME	7:45AM
LABORATORY TESTING	☐ YES ☒ NO		CLEANLINESS ACHIEVED	AIR TEMP	81°F
TYPE OF ABRASIVE	STEEL GRIT		PROFILE REQUIRED	RELATIVE HUMIDITY	78%
NOZZLE PSI	☐ N/A 100 PSI ☒		PROFILE ACHIEVED	STEEL	82°F
BLOTTER TEST	☒ PASS		SSPC-VIS 1 USED	DEW POINT	73°F
DUST FREE SURFACES	☒ YES ☐ NO		VISUAL INSPECTION	STAGE OF OPERATION	PRE-BLAST
COATING APPLICATION					
COAT 1 ST	PRODUCT NAME: PIPECLAD 2000		BATCH: CH1416SW	MIN MILS: 14	TARGET PRE COAT TEMP: 450 DEG F
FINAL TESTING					
HVST @125V/MIL; 2000V MIN PASS ☒		ADHESION KNIFE TEST: PASS ☒		DFT TEST: PASS ☒	
ALL REQUIREMENTS MET: PASS ☒					
KI#	DESCRIPTION	MIL AVERAGE	PRESS O FILMS	KI#	DESCRIPTION
1-1	8" X .322 30DEG BEND	27	Testex PRESS-O-FILM™ HT X-Coarse 1.5 to 4.5 mil 30 to 115 µm Made in USA 3.4	2-2	8" X .322 45DEG BEND
1-2	8" X .322 30DEG BEND	26			
1-3	8" X .322 30DEG BEND	28	Testex PRESS-O-FILM™ HT X-Coarse 1.5 to 4.5 mil 30 to 115 µm Made in USA 3.4		
1-4	8" X .322 30DEG BEND	25			
2-1	8" X .322 45DEG BEND	27	Testex PRESS-O-FILM™ HT X-Coarse 1.5 to 4.5 mil 30 to 115 µm Made in USA 3.4		

VERIFIED BY KEITH:



Production Bend Hardness Report

Customer:	DNOW	Report No.:	109149 H 1
Customer PO No.:	662974020	Code of Fabrication:	
TTB Job No.:	26H109149	ASME B16.49 2023	

Mother Pipe Information

Pipe Size (in): 8.625 x 0.322

Process Information

Grade: API-5L-X52 PSL 2

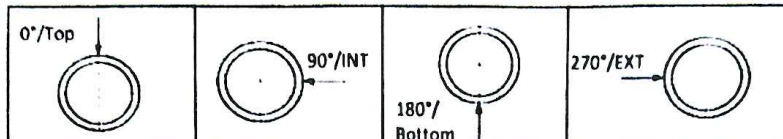
Instrument Model No.:

UCI-3000H

Serial No. MK8821415112 1754

Probe Serial No. MK8823417572

Key:



T = Top (Weld if Welded Material)

H = Weld Heat Affected Zone

I = Intrados of Bend

E = Extrados of Bend

TAN = Tangent of Bend

STZ = Start Transition Zone

FTZ = Finish Transition Zone

3/4 = 3/4 Point of the Bend Arc

Note 1. Weld & Weld HAZ readings are only required on bends made from welded mother pipe.

Note 2. Readings at 3/4 bend arc locations are only required for production bends at least 30° greater than the QB angle.

Hardness Readings

Hardness Readings			Min Allowed Values:			137	Max Allowed Values:			238		3/4= 3/4 Point of the Bend Arc			
Bend ID No.	TAN T 0°	TAN H 0°	TAN 270°	STZ E 270°	STZ I 90°	MID T 0°	MID H 0°	MID E 270°	MID I 90°	FTZ E 270°	FTZ I 90°	3/4 T 0°	3/4 H 0°	3/4 EXT 270°	3/4 INT 90°
109149-1-1	194	189	199	188	182	189	180	181	186	189	185	N/A	N/A	N/A	N/A
109149-1-2	188	179	188	180	190	186	175	182	192	186	189	N/A	N/A	N/A	N/A
109149-1-3	195	188	183	179	180	203	190	182	177	183	178	N/A	N/A	N/A	N/A
109149-1-4	189	181	186	184	181	183	181	190	183	183	184	N/A	N/A	N/A	N/A
109149-2-1	192	187	190	186	184	197	190	183	183	188	187	N/A	N/A	N/A	N/A
109149-2-2	192	187	181	185	183	196	194	185	182	190	186	N/A	N/A	N/A	N/A

READINGS: ☒ MEET ACCEPTANCE CRITERIA
☐ DO NOT MEET ACCEPTANCE CRITERIA

PERFORMED BY: JV

DATE: 6/12/2026

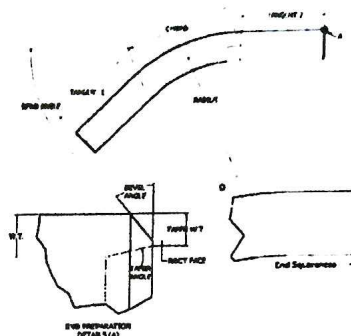
 Dimensional-Visual Inspection Report					
Customer:	INNOV	Pipe Size (in)	8.625 x 0.122	Report No.:	109149 157 2 2
Customer PO No.:	667974020	Grade:	API 5L X 62 P 5 2	Bend ID No.:	109149 2 2
TTR Job No.:	261109149	Heat No.:	R416/40	Code of Fabrication:	A777 B15 49 1573

Physical Dimensions

Dimension	Target	Tolerance	Actual
Tangent 1 (in)	24 MIN	MIN	26.13
Tangent 2 (in)	24 MIN	MIN	28.38
Chord (in)	30.6	n/a	30.5
Angle (deg)	45.0	+/- 0.5°	44.8
Radius, R (in)	40	+/- 1%	40.0
Performed By:	MF	Date:	6/15/2026

End Prep Dimensions

Dimension	Target	Tolerance	Actual	
Bevel Angle (deg)	37.5	+/- 2.5°	40.00	END 1
Root Face (in)	0.06	+/- 0.03 in	0.060	
Min ID	7.981	+/- 0.1 in	7.937	
Max ID	7.981	+/- 0.1 in	7.937	
Max End Sq. (in)	0	+/- 0.09 in	0.060	
Bevel Angle (deg)	37.5	+/- 2.5°	40.00	END 2
Root Face (in)	0.06	+/- 0.03 in	0.060	
Min ID	7.981	+/- 0.1 in	7.937	
Max ID	7.981	+/- 0.1 in	7.937	
Max End Sq. (in)	0	+/- 0.09 in	0.060	
Performed By:	MF	Date:	6/15/2026	



Ovality [(O.D. max - O.D. min)/O.D. nominal] x 100

Quadrant	Limit	Min OD	Max OD	% Ovality
End 1	1.0%	8.585	8.602	0.2%
Start of Bend	1.0%	8.606	8.619	0.2%
1/4 of Bend Arc	1.0%	8.598	8.637	0.5%
1/2 of Bend Arc	1.0%	8.591	8.640	0.6%
3/4 of Bend Arc	1.0%	8.596	8.650	0.6%
Stop of Bend	1.0%	8.610	8.636	0.3%
End 2	1.0%	8.574	8.603	0.3%
Performed By:	MF	Date:	6/15/2026	

Wall Thinning of Extrados

Quadrant	Limit	Nominal Wall Thickness	Extrados Wall Thickness	% Wall Thinning
Start of Bend	10.0%	0.322	0.311	3.4%
1/4 of Bend Arc	10.0%	0.322	0.300	6.8%
1/2 of Bend Arc	10.0%	0.322	0.302	6.2%
3/4 of Bend Arc	10.0%	0.322	0.300	6.8%
Stop of Bend	10.0%	0.322	0.309	4.0%
Performed By:	EB	Date:	6/12/2026	

ID Gauge Drift/Pig Inspection

Minimum Gauge Diameter =	97.0%	of Nominal ID, or	7.742 in
Gauge Length Required =	7.981 in		
Gauge passes through the entire bend without power equipment assistance			
Performed By:	MF	Date:	6/15/2026

Visual Inspection

All accessible surfaces found to be free of visible injurious defects.	
Bend is stenciled with correct information.	
Performed By:	MF Date: 6/15/2026

READINGS: ☒ MEET ACCEPTANCE CRITERIA
☐ DO NOT MEET ACCEPTANCE CRITERIA